Course: B.Tech
Code: 104506 **Subject: Probability Theory and Stochastic Process** **Full Marks: 70**

Instructions:-

- *(i) The marks are indicated in the right-hand margin.*
- *(ii) There are NINE questions in this paper.*
- *(iii) Attempt FIVE questions in all.*
- *(iv) Question No. 1 is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A fair die is rolled twice. What is the probability that the sum is 7?
(i) 1/12
(ii) 1/6
(iii) 1/3
(iv) 1/2

(b) If A and B are any two events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$ and $P(A^c) = 2/3$, then $P(A^c/B)$ is equal to
(i) 5/8
(ii) 3/8
(iii) 3/4
(iv) 1/8

(c) A discrete random variable X has values 1, 2, and 3 with probabilities 0.2, 0.5, and 0.3 respectively. The mean $E(X)$ is equal to
(i) 1.8
(ii) 2.1
(iii) 2.5
(iv) 3

(d) If the probability density function is $f(x) = 2x$ for $0 < x < 1$, then $P(X < 0.5)$ is
(i) 0.25
(ii) 0.5
(iii) 0.75
(iv) 1

(e) For a binomial distribution with $n = 4$ and $p = 0.5$, the mean is
(i) 1
(ii) 3
(iii) 2
(iv) 4

(f) Two coins are tossed simultaneously. What is the probability of getting exactly one head?
(i) 1/4
(ii) 1/2
(iii) ¾
(iv) 1

(g) Central limit theorem applies when
(i) n small
(ii) $n \to \infty$
(iii) discrete only
(iv) continuous only

(h) If a random variable has mean 10 and variance 4, then by Chebyshev inequality, $P(|X - 10| \geq 4) \leq \cdots$
(i) 0.5
(ii) 0.75
(iii) 0.25
(iv) 1

(i) A Markov chain has transition probabilities whose row sums are equal to
(i) 0
(ii) 1
(iii) 2
(iv) Number of States

(j) Ergodic process
(i) Time avg = ensemble avg
(ii) Time avg ≠ ensemble avg
(iii) Deterministic
(iv) None

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Q.2 (a) State and prove Baye's theorem. [7]

(b) When two dice are thrown, find the probability of getting the sums of 10 or 11? [7]

Q.3 (a) Give the classical definition of probability. Discuss joint probability and conditional probability with an example. [7]

(b) A random variable X has the following probability distribution [7]

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

Find (i) k, (ii) $P(X < 6)$, (iii) mean, and (iv) variance.

Q.4 (a) A continuous random variable X has probability density function $f(x) = \begin{cases} a(2x - x^2), 0 \le x \le 2 \\ 0 \quad\quad\quad , otherwise \end{cases}$. Find (i) a, (ii) $P(X > 1)$ and (iii) mean. [7]

(b) Suppose the random variable X has a finite second moment, then for any constant $c > 0$, prove that $P(| X | \geq c) \leq \frac{E(X^2)}{c^2}$. [7]

Q.5 (a) The joint probability density function of the random variables X and Y is given as $f(x,y) = \begin{cases} 8xy, \ 0 \le x \le 1, 0 \le y \le x \\ 0, \quad\quad\quad otherwise \end{cases}$. Find the marginal distribution of X and Y and conditional distribution of Y for X. [7]

(b) Compare Markov inequality, Chebyshev inequality, and Chernoff bound with respect to assumptions, applicability, and tightness of bounds. [7]

Q.6 (a) State the Strong Law of Large Numbers and explain how it differs from Weak Law of Large Numbers. [7]

(b) Briefly introduce the concept of random process and categorize its classifications with one example to each one. [7]

Q.7 (a) A random process is defined as $X(t) = A\,sin(\omega t + \theta)$, where A is a constant and θ is a random variable distributed over $(\pi, -\pi)$, check $X(t)$ is stationary. [7]

(b) Population mean = 50, variance = 25, n=100. Use Central Limit Theorem to find $P(45 \leq X \leq 55)$. [7]

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Q.8 (a) Classify the states of the Markov chain with following transition probabilities [7]

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$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{1}{3} \end{bmatrix}$$

(b) Derive the relationship between the power spectral density of the input and output random processes of an LTI system. [7]

Q.9 Write short note on *any two* of the following: [7 x 2 = 14]

(a) Definition and types of random variables
(b) Band-pass random processes
(c) Conditional distribution functions and their types
(d) Stationary processes and ergodicity

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